

## Lecture 9

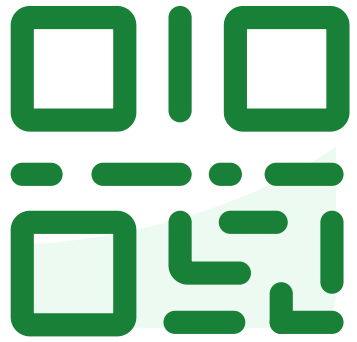
# Regression (Cont.) and Bias-Variance Trade-off

More on probabilistic view of regression and bias-variance trade-off

**EECS 189/289, Fall 2025 @ UC Berkeley**

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# Roadmap

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off





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# MLE Recap: Least Squares as Maximum Likelihood

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
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- Bias-Variance Trade-off



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# Least Squares $\equiv$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

The optimum value of  $\mathbf{w}$  in the above equation equals the optimum value of the log of the function (because log is a monotonically increasing function)

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = \sum_{n=1}^N \ln(\mathcal{N}(t_n | \overset{\mu}{\mathbf{w}^T \phi(\mathbf{x}_n)}, \sigma^2))$$

Writing the equation for a Normal distribution

$$= \sum_{n=1}^N \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2}\right)$$

Applying  $\log a \times b = \log a + \log b$

$$= \sum_{n=1}^N \left( \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}(t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \right)$$

Simplify and separate two terms

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{\sigma^2} E_D(\mathbf{w})$$



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# Least Squares $\equiv$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2} E_D(\mathbf{w})$$

$$\nabla_{\mathbf{w}} \ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{1}{\sigma^2} \nabla_{\mathbf{w}} E_D(\mathbf{w}) \longrightarrow \mathbf{w}_{ML} = (\Phi(\mathbb{X})^T \Phi(\mathbb{X}))^{-1} \Phi(\mathbb{X})^T \mathbf{t}$$

$$\Phi(\mathbb{X}) = \begin{bmatrix} \phi(\mathbf{x}_1)^T \\ \phi(\mathbf{x}_2)^T \\ \vdots \\ \phi(\mathbf{x}_N)^T \end{bmatrix}$$

The least-squares solution is the **MLE** under Gaussian noise.

# Choosing Different Noise Models

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off

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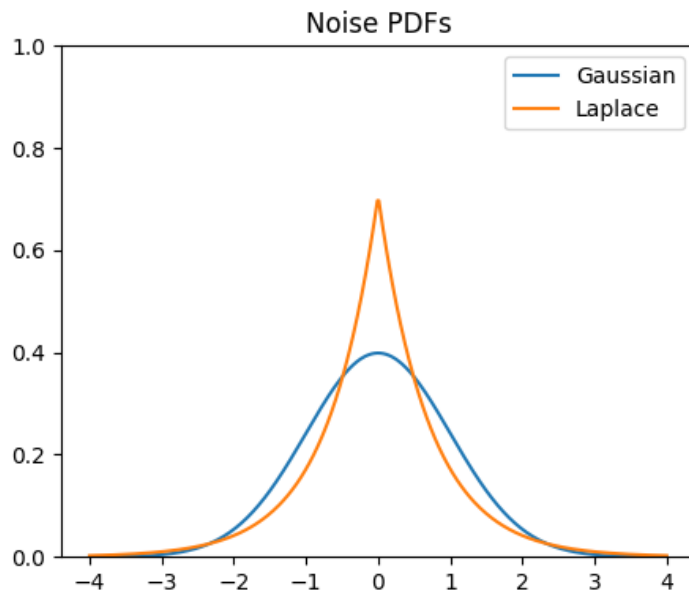
# Noise Model $\Leftrightarrow$ Error Function

## Zero-mean Gaussian Noise

$$p(\epsilon) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$$

$$-\log p(\epsilon) = \frac{\epsilon^2}{2\sigma^2} + \text{const}$$

$\Rightarrow$  Sum of the squares error

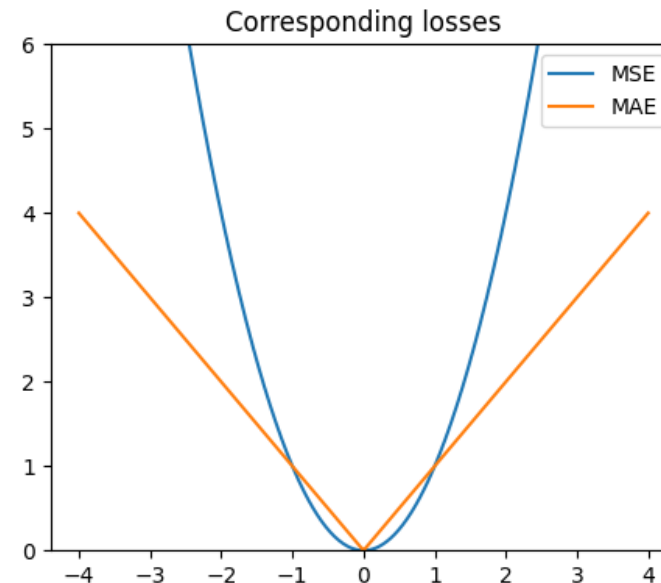


## Zero-mean Laplacian Noise

$$p(\epsilon) = \frac{1}{2b} \exp\left(-\frac{|\epsilon|}{b}\right)$$

$$-\log p(\epsilon) = \frac{|\epsilon|}{b} + \text{const}$$

$\Rightarrow$  Sum of the Absolute error







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# Prior Beliefs

- MLE Recap: Least Squares as Maximum Likelihood
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# Recall: Beliefs and Priors

In the coin flip example, assume the true probability of head is  $\theta$ .

- **Frequentist:** Use data only (MLE).
- **Bayesian:** Combine data with belief.

$$\hat{\theta} = \frac{\#heads}{\#flips}$$

- Prior belief:  $h_0 = t_0 = 5$
- See HH:
  - Posterior  $p(\theta|HH) \propto p(\theta) \times \text{likelihood}$

Flip twice, see HH, then:

$$\hat{\theta}_{MLE} = \frac{2}{2} = 1$$

Strong prior ensures alignment with beliefs

Chases the data  
 $p(HH|\theta) = \theta^2$

Overconfident with  $n = 2!!!$

$$\hat{\theta}_{posterior\ mean} = \frac{5 + 2}{10 + 2} = 0.58$$

0 ——— 0.5 (prior) ——— 0.58 (Posterior mean) ——— 1.0 (MLE)

# What About Regression? MLE for Weights: No Prior



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- In regression, MLE finds weights  $\mathbf{w}$  by maximizing likelihood of data under noise model.
- **Problem:** with few/noisy data points  $\rightarrow$  can fit extreme weights.
- No mechanism to encode “weights should be small/reasonable.”
- Why small?
  - With inputs in SD units we don't expect huge effects unless data is persuasive.
  - Shrinking  $\mathbf{w}$  lowers prediction variance, especially when features are correlated.



# Belief about Parameters

- Just like coins: we can encode a prior belief about weights  $\mathbf{w}$ .
- Common prior:  $w_j \sim \mathcal{N}(0, \sigma_w^2)$ 
  - Centered at 0  $\rightarrow$  no strong effect unless data supports it.
  - Variance  $\sigma_w^2$  controls how strongly we expect small weights.
- This prior = belief that simpler models are preferred.

Likelihood function

$$p(\mathbf{t} | \mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

Prior (small, centered)

$$p(\mathbf{w} | \tau^2) = \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}$$

Posterior (from Bayes rule) up to a constant:

$$p(\mathbf{w} | \mathbf{t}) \propto$$



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# What Is MAP?

Bayes Rule

$$p(\mathbf{w} \mid \mathbf{t}) \propto \underbrace{\prod_{n=1}^N \mathcal{N}(t_n \mid \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)}_{\text{Likelihood}} \underbrace{\frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}}_{\text{Prior}}$$

Maximum A Posterior (MAP):

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} \mid \mathbf{t}) = \underset{\mathbf{w}}{\operatorname{argmax}} [p(\mathbf{t} \mid \mathbf{w}) \times p(\mathbf{w})]$$



# What Is MAP?

Maximum A Posterior (MAP):

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmax}} p(w | t) = \underset{w}{\operatorname{argmax}} [p(t | w) \times p(w)]$$

Taking  $-\log$  of the objective turns maximization problem to a minimization problem:

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} [-\ln p(t | w) - \ln p(w)]$$



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# Plugging in Distributions

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[ \underbrace{-\ln p(\mathbf{t} | w)}_{\text{data fit}} - \underbrace{\ln p(w)}_{\text{prior}} \right]$$

$$\begin{aligned} & -\ln \prod_{n=1}^N \mathcal{N}(t_n | w^T \phi(\mathbf{x}_n), \sigma^2) \quad -\ln \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|w\|^2}{2\tau^2}} \\ &= \boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(\mathbf{x}), w))^2} = \boxed{\frac{\|w\|^2}{2\tau^2}} + \text{const} \end{aligned}$$

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[ \boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(\mathbf{x}), w))^2} + \boxed{\frac{\|w\|^2}{2\tau^2}} \right]$$



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# Does This Look Like a Ridge Regression?

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[ \frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(x), w))^2 + \frac{\|w\|^2}{2\tau^2} \right]$$

Multiply by  $\sigma^2$

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[ \underbrace{\frac{1}{2} \sum_n (t_n - y(\phi(x), w))^2}_{E_D(w)} + \underbrace{\frac{\sigma^2 \|w\|^2}{2\tau^2}}_{E_w(w)} \right]$$

$\lambda$

Least-Squares + Gaussian Prior  $\Rightarrow$  Ridge Regression





# What is the relationship between MAP and ridge regression?

# Bias-Variance Trade-off

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off





# What are some fundamental challenges in machine learning?

# Fundamental Challenges in Learning?



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- ***Fit the Data***
  - Provide an explanation for what we observe
- ***Generalize to the World***
  - Predict the future
  - Explain the unobserved



Is this cat grumpy or are we  
overfitting to human faces?



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# Fundamental Challenges in Learning?

- **Bias:** The expected deviation between the predicted value and the true value.
- **Variance:** Two sources
  - **Noise:** The variability of the random noise in the process we are trying to model.
  - **Model Variance:** The variability in the predicted value across different training datasets.

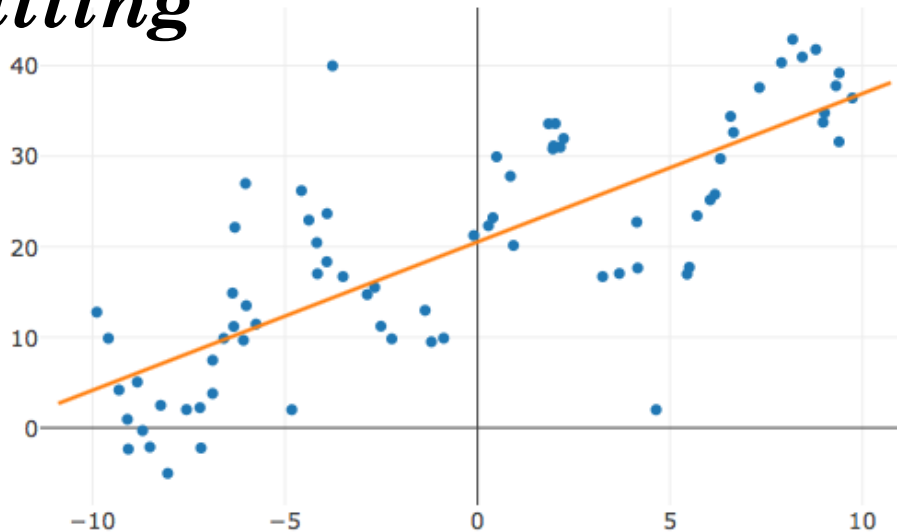
# Bias



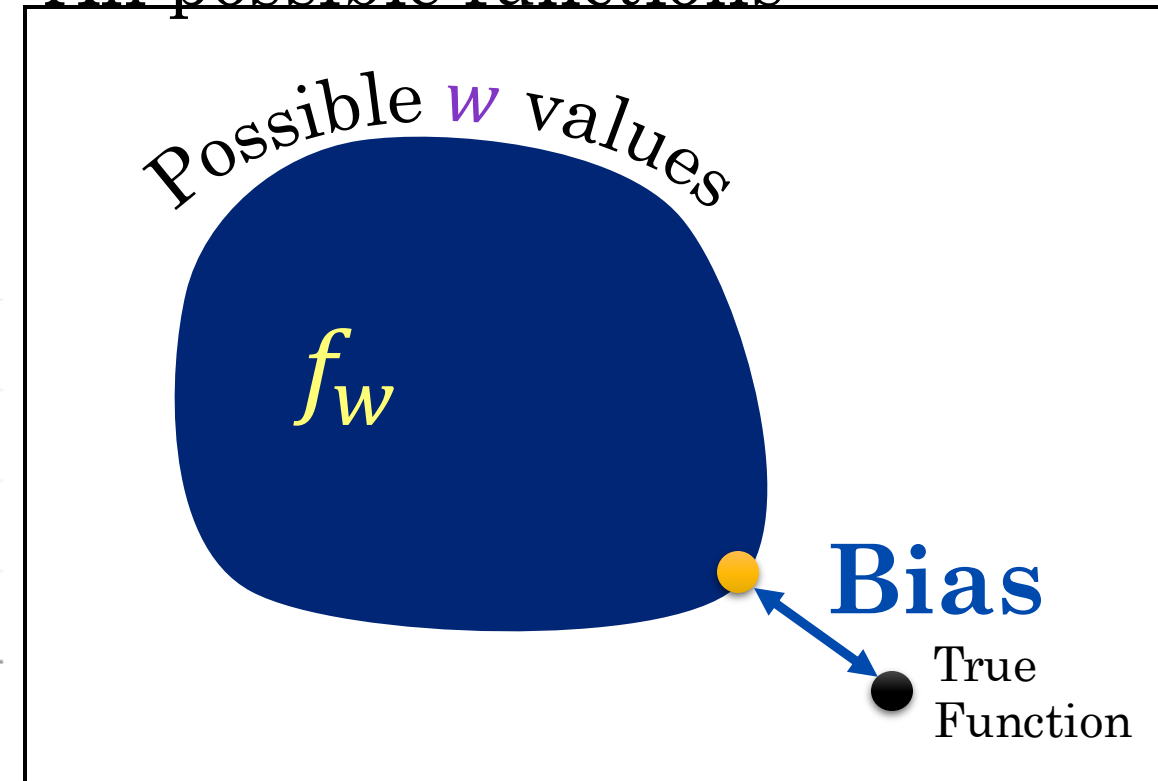
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*The expected deviation between the predicted value and the true value*

- Depends on both the:
  - Choice of  $f$
  - Learning procedure
- ***Under-fitting***



All possible functions



# Noise

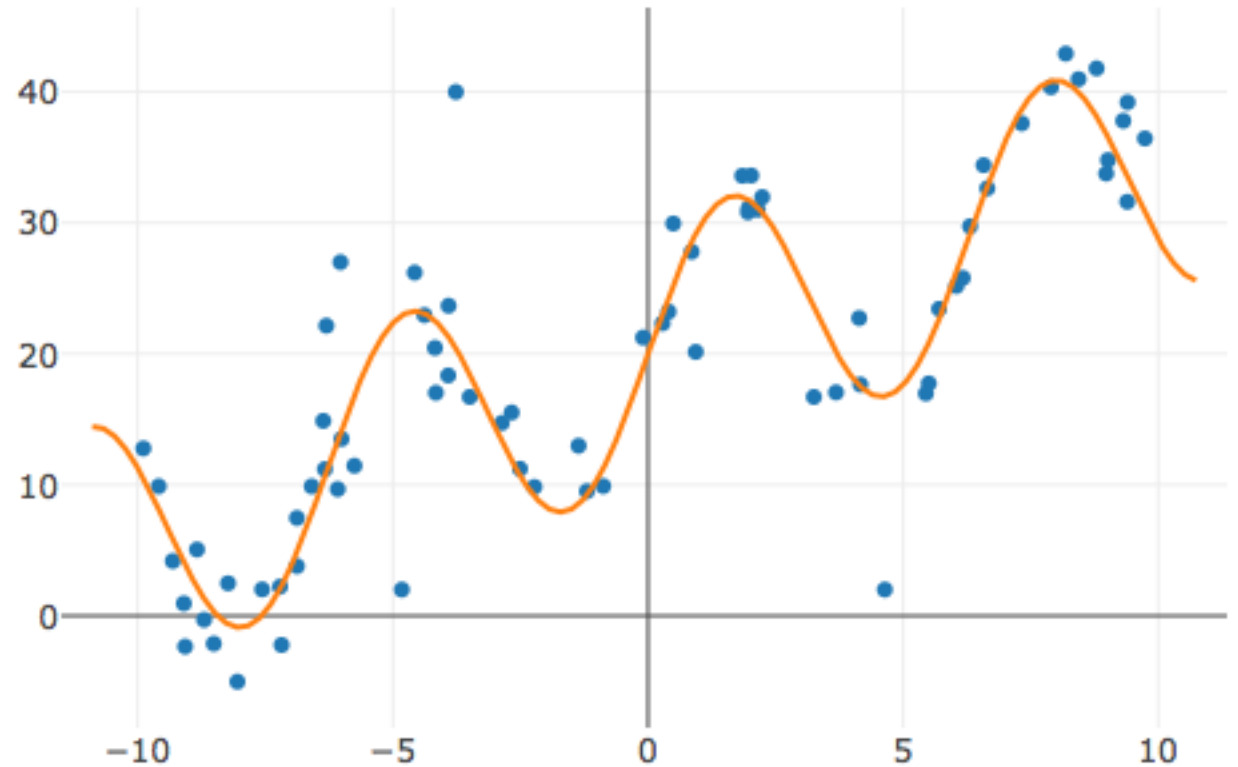


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The variability of the random noise in the process we are trying to model.

- Measurement variability
- Stochasticity
- Missing information

**Beyond our control  
(usually)**



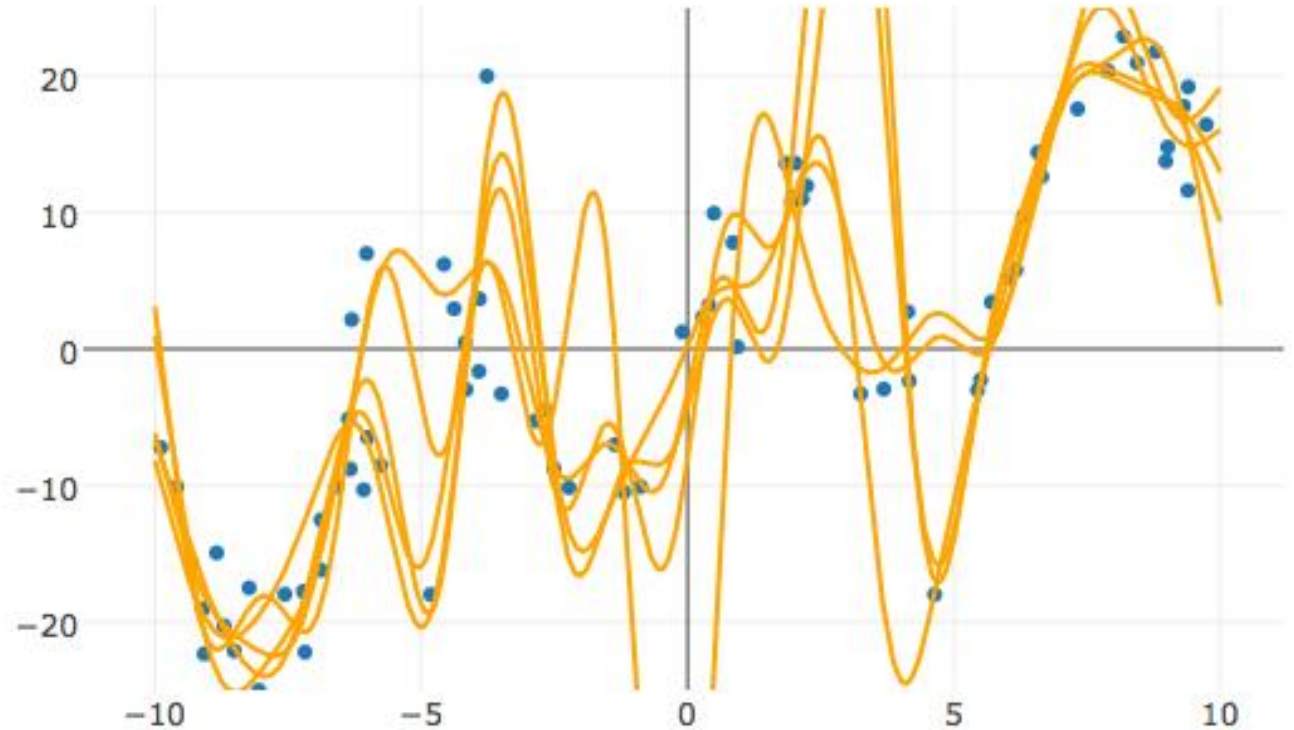


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# Model Variance

*Variability in the predicted value across different training datasets.*

- Sensitivity to variation in the training data
- Poor generalization
- **Overfitting**







# Which of the following models would have high bias?



**Which of the following models  
would have high variance?**

# Analysis of Squared Error



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- For the test point  $\mathbf{x}$  the expected error:
  - *Random variables are red*

Assume noisy observations  
→  $t$  is a random variable

True Function

$$t = h(\mathbf{x}) + \epsilon$$

Noise term:

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

Can be any  
parametric function

Assume *training data* is  
random  
→  $w$  is a random variable

# Analysis of Squared Error



Goal:

$$E[(t - f_w(x))^2] =$$

**“Noise” + (Bias)<sup>2</sup> + Model Variance**



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$$E[(t - f_w(\mathbf{x}))^2] = E[(t - \underbrace{h(\mathbf{x}) + h(\mathbf{x})}_{\text{Subtracting and adding } h(\mathbf{x})} - f_w(\mathbf{x}))^2]$$

Subtracting and adding  $h(\mathbf{x})$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2] = E[\underbrace{(t - h(\mathbf{x}))}_a + \underbrace{h(\mathbf{x}) - f_w(\mathbf{x}))}_b]^2]$$

Expanding in terms of  $a$  and  $b$ :  $(a + b)^2 = a^2 + b^2 + 2ab$

$$\begin{aligned} &= E[(t - h(\mathbf{x}))^2] + E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2] \\ &\quad + 2 E[\underbrace{(t - h(\mathbf{x}))}_\epsilon \underbrace{(h(\mathbf{x}) - f_w(\mathbf{x}))}_{2ab}] \\ &\quad + 2 E[\epsilon (h(\mathbf{x}) - f_w(\mathbf{x}))] \end{aligned}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2] = E[\underbrace{(t - h(\mathbf{x}))}_a + \underbrace{h(\mathbf{x}) - f_w(\mathbf{x}))}_b]^2]$$

Expanding in terms of  $a$  and  $b$ :  $(a + b)^2 = a^2 + b^2 + 2ab$

$$= E[(t - h(\mathbf{x}))^2] + E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2] + 2 E[\epsilon (h(\mathbf{x}) - f_w(\mathbf{x}))]$$

Independence of  $\epsilon$  and  $w$

$$+ 2 \underbrace{E[\epsilon]}_0 E[(h(\mathbf{x}) - f_w(\mathbf{x}))]$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$\begin{aligned} & \mathbb{E}[(t - f_w(\mathbf{x}))^2] \\ &= \mathbb{E}[(t - h(\mathbf{x}))^2] + \mathbb{E}[(h(\mathbf{x}) - f_w(\mathbf{x}))^2] = \sigma^2 \end{aligned}$$

Obs. Value      True Value

True Value      Pred. Value

*“Noise”* Term

Model  
Estimation  
Error

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$\mathbb{E}(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$





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$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model  
Estimation  
Error

We need to calculate  
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model  
Estimation  
Error

We need to calculate  
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] = \text{Next we will show...}$$

$$\underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2}_{(\text{Bias})^2} + \underbrace{\mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]}_{\text{Model Variance}}$$

➤ How?

➤ Adding and subtracting what?



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] =$$

$$\mathbb{E} \left[ \underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_a + \underbrace{\left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)}_b \right]^2$$

Expanding in terms of  $a$  and  $b$ :  $(a + b)^2 = a^2 + b^2 + 2ab$

$$\begin{aligned} & \mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] + \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ & 2\mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right] \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] &+ \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ 2\mathbb{E} \left[ \underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_{\text{constant}} \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right] & \\ \underbrace{2 \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} & \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] &+ \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ 2 \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) &\underbrace{\mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} \\ \underbrace{2 \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_{\text{constant}} &\underbrace{\left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_0 \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \underbrace{\mathbb{E} \left[ \left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right]}_{\text{constant}} &+ \mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] \\ (h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})])^2 &+ \end{aligned}$$



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$$\mathbb{E} \left[ \left( h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] =$$
$$\underbrace{\left( h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2}_{(\text{Bias})^2} + \underbrace{\mathbb{E} \left[ \left( \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]}_{\text{Model Variance}}$$



$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model  
Estimation  
Error

We now have  
calculated this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= \sigma^2$$

$$= E[(t - h(\mathbf{x}))^2] + \text{“Noise” Term}$$

$$(h(\mathbf{x}) - E[f_w(\mathbf{x})])^2 + \text{(Bias)}^2$$

$$E[(E[f_w(\mathbf{x})] - f_w(\mathbf{x}))^2] \text{ Model Variance}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



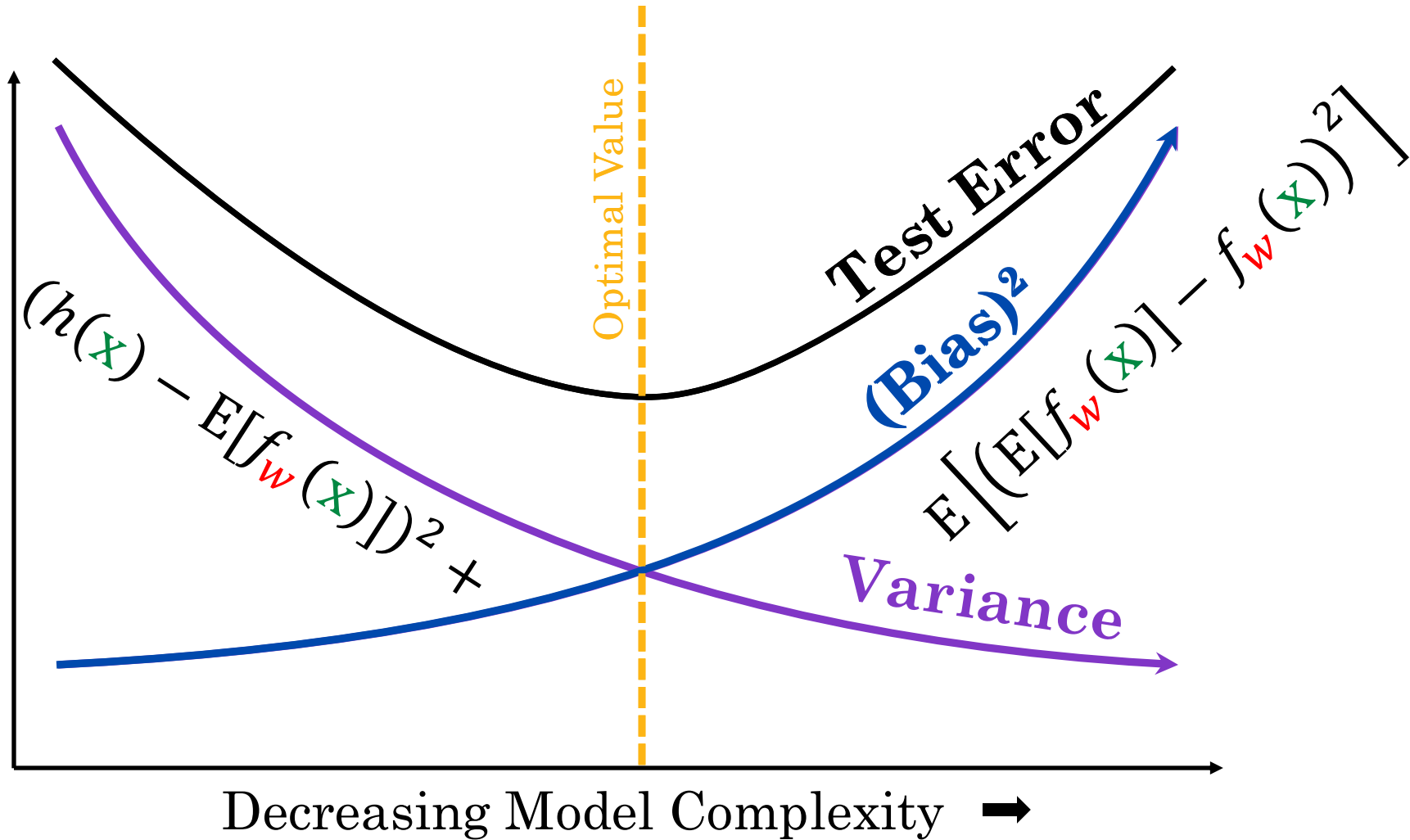
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$$\begin{aligned} & \mathbb{E}[(t - f_w(\mathbf{x}))^2] \\ &= \mathbb{E}[(t - h(\mathbf{x}))^2] + \quad \text{“Noise” Term} \\ & \quad (h(\mathbf{x}) - \mathbb{E}[f_w(\mathbf{x})])^2 + \quad \text{(Bias)}^2 \\ & \quad \mathbb{E}[(\mathbb{E}[f_w(\mathbf{x})] - f_w(\mathbf{x}))^2] \quad \text{Model} \\ & \quad \text{Variance} \end{aligned}$$

# Bias Variance Plot



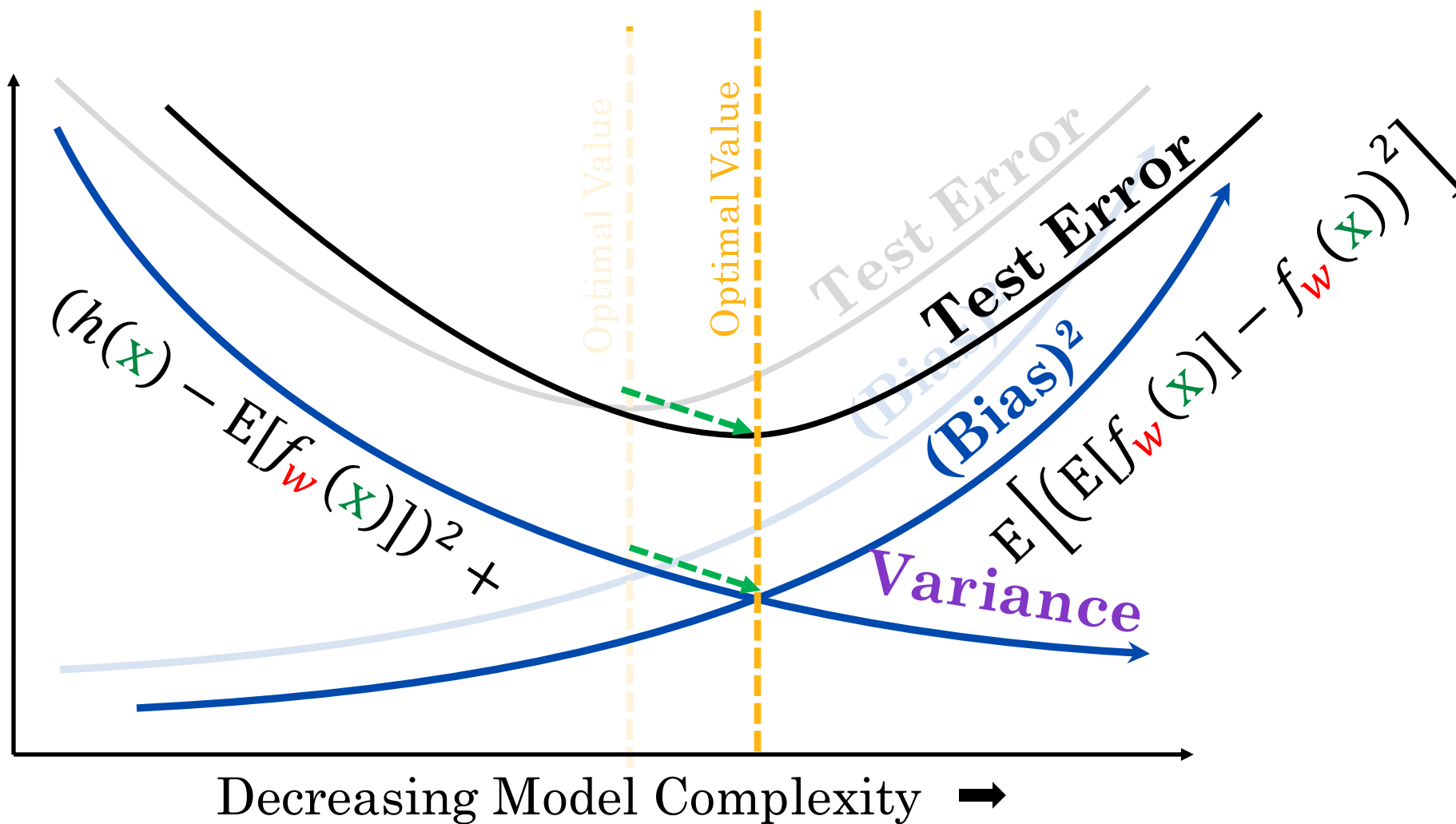
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# More Data Supports More Complexity



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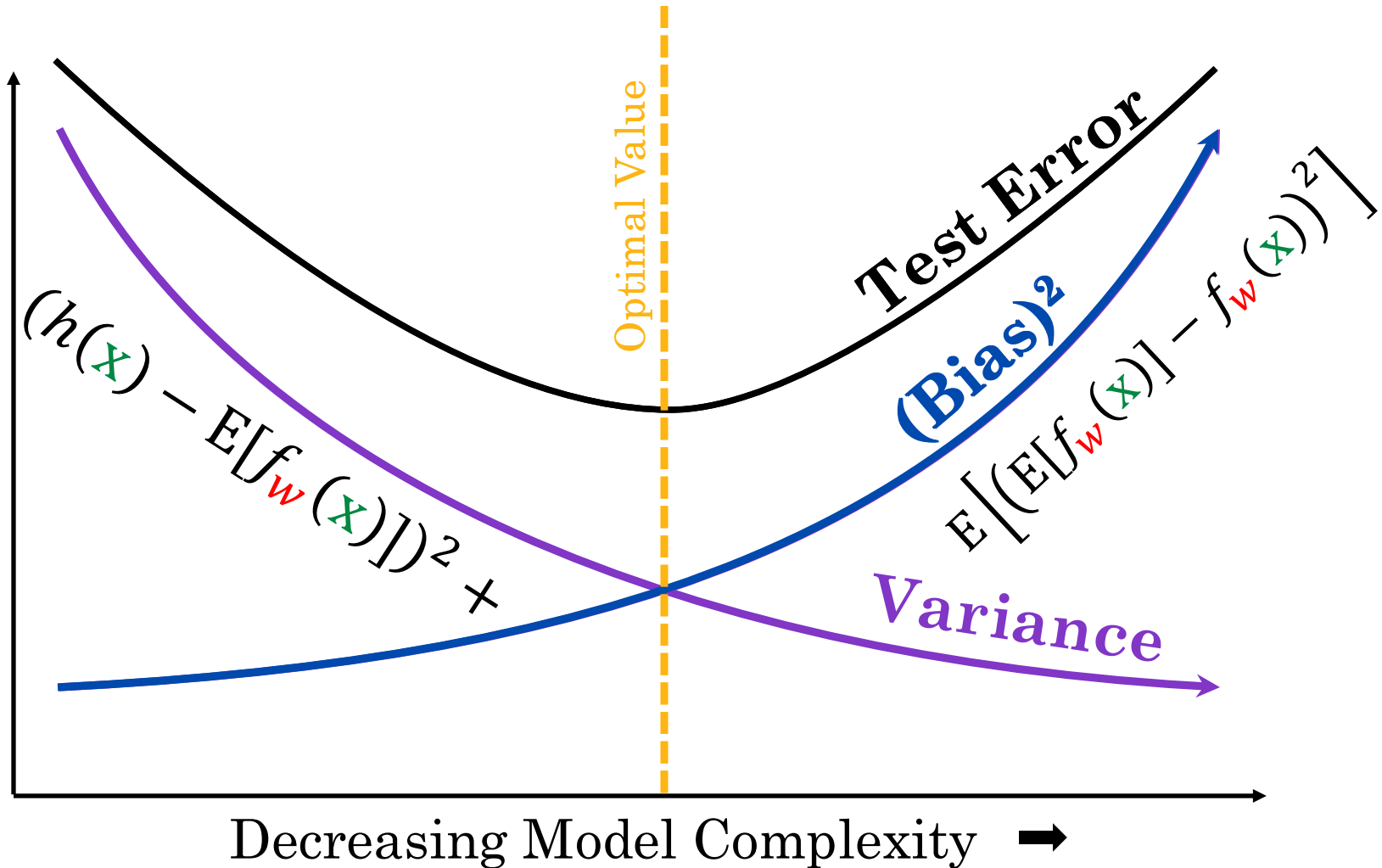


# How Do We Control Model Complexity?

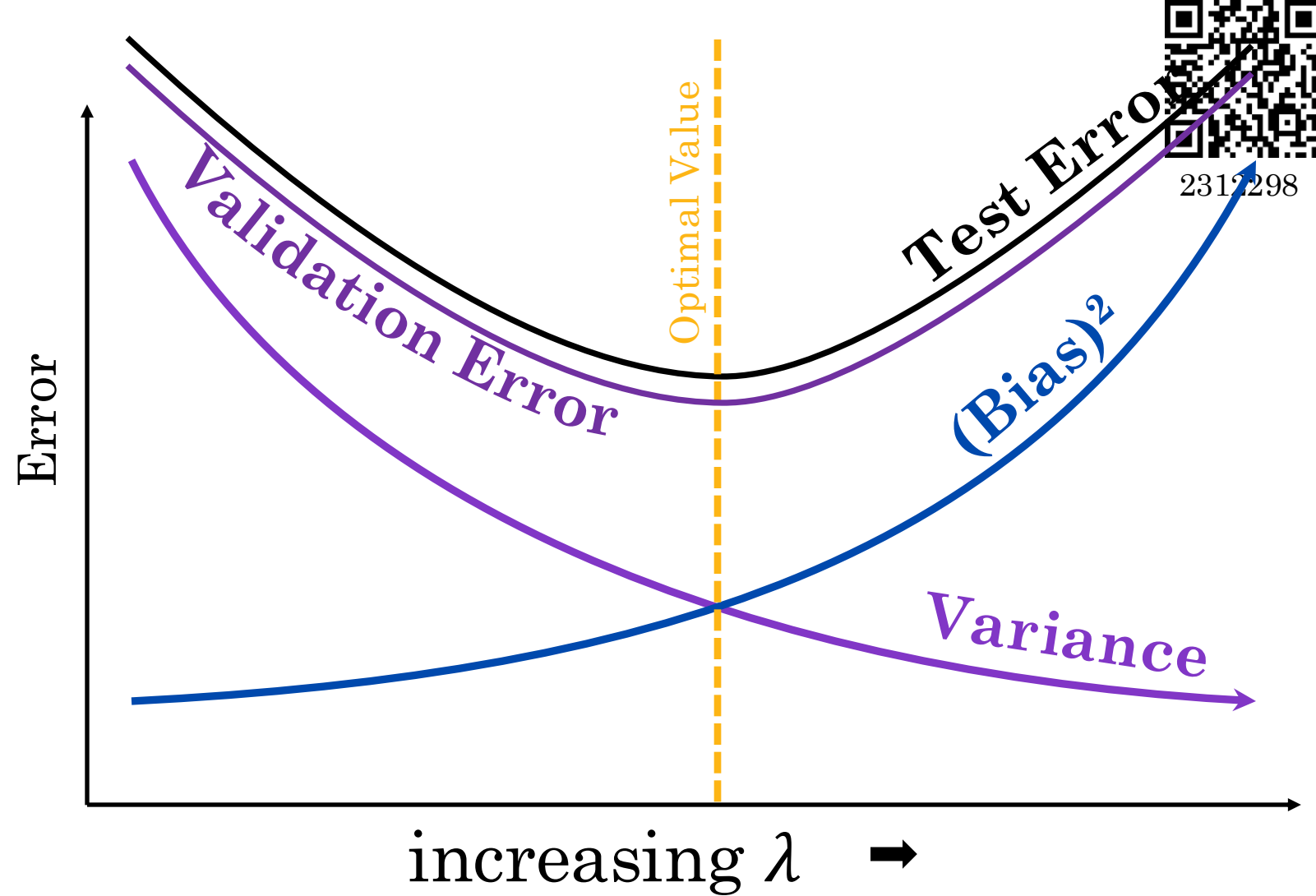


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- So far:
  - Number of features
  - Choices of features
  - **Regularization**



# Determining the Optimal $\lambda$



- Value of  $\lambda$  determines bias-variance tradeoff
  - Larger values  $\rightarrow$  more regularization  $\rightarrow$  more bias  $\rightarrow$  less variance
- Determined through validation



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# Dataset Example

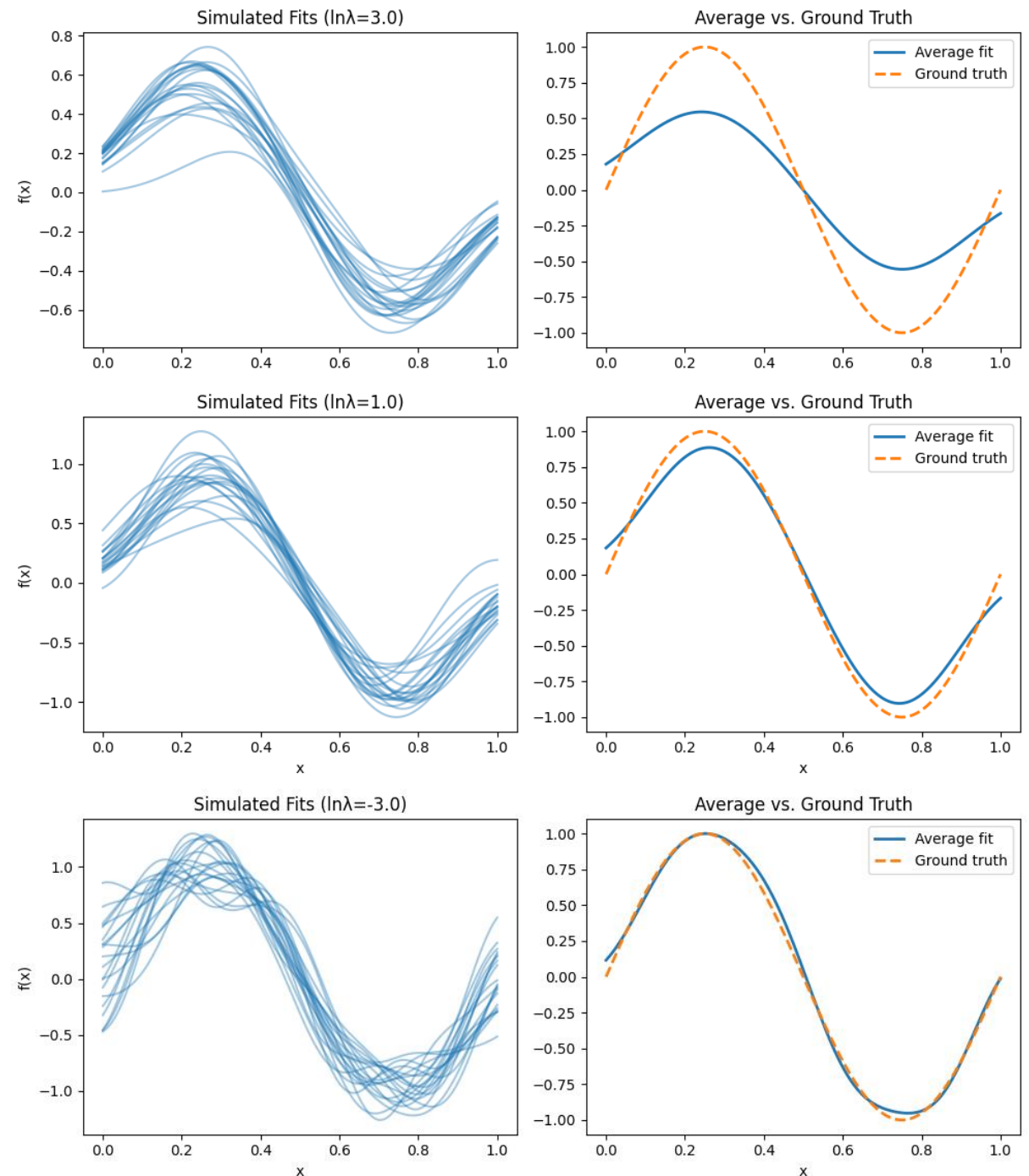
**Data generation function:**

$$h(x) = \sin(2\pi x), x \sim \text{Uniform}(0,1)$$

**Generate**  $L$  independent datasets ( $L = 100$ ), each with  $N = 25$  noisy observations.

$$t_n = h(x_n) + \epsilon_n, \\ \epsilon_n \sim \mathcal{N}(0, 0.32)$$

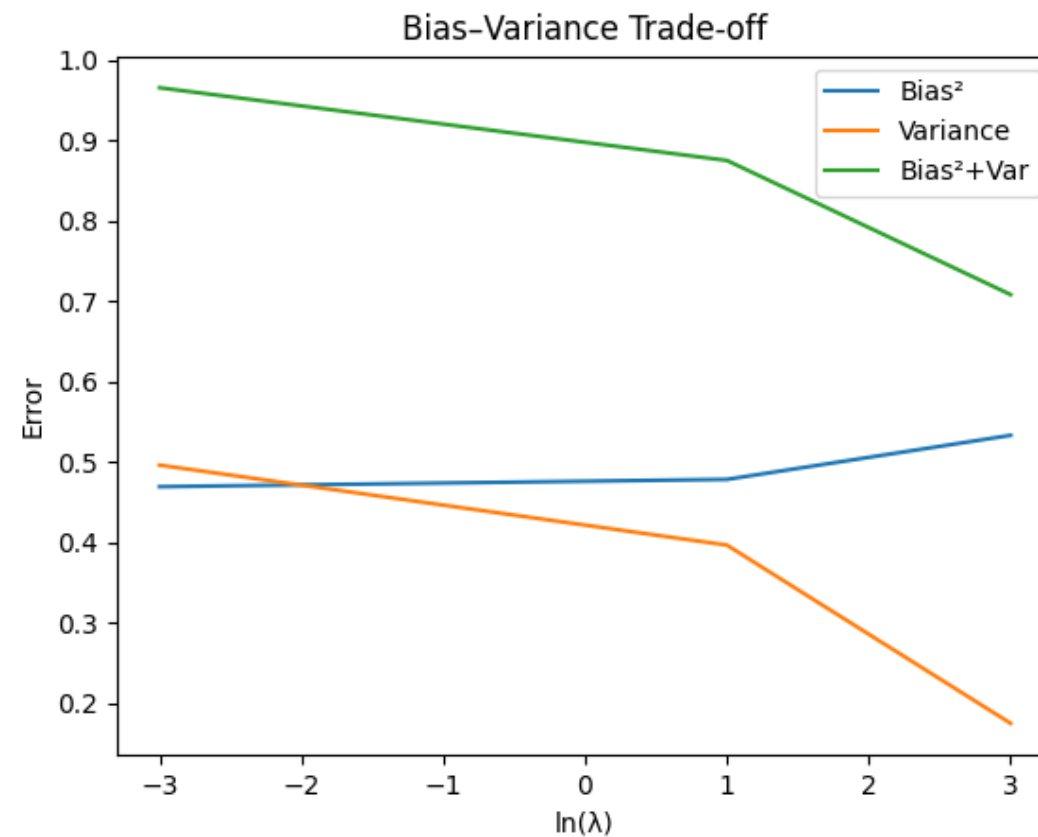
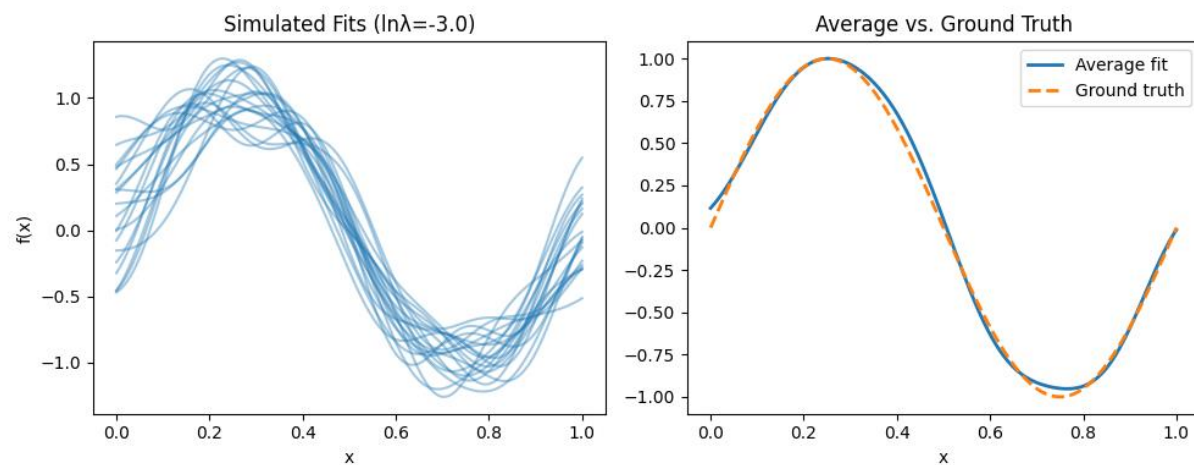
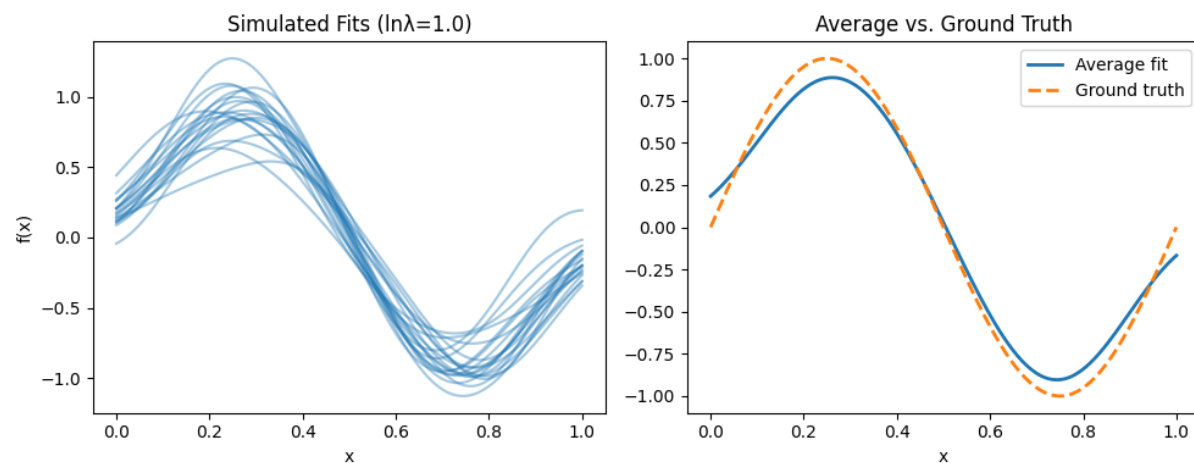
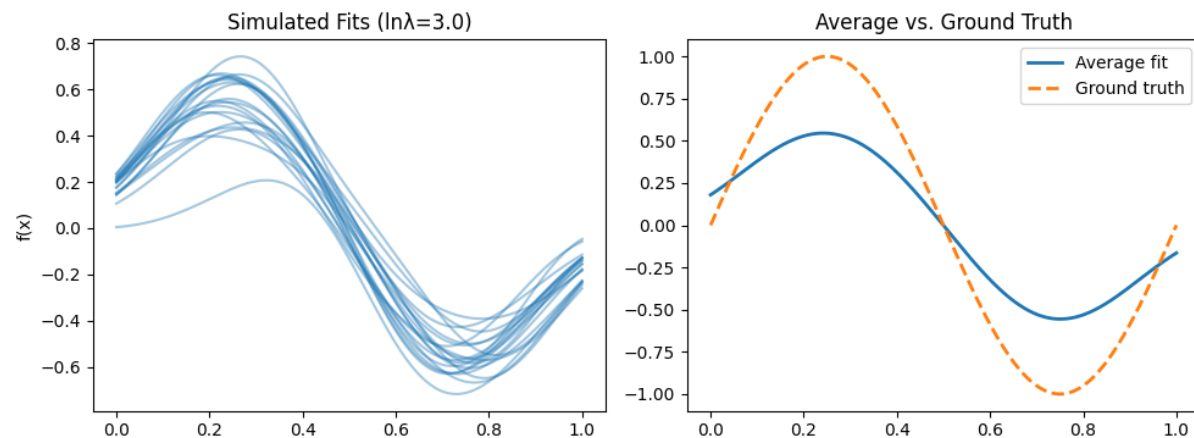
**Model:** linear combination of  $M = 24$  Gaussian basis functions plus bias.







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**Select options below.**



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## Lecture 9

# Regression (Cont.) and Bias-Variance Trade-off

**Credit:** Joseph E. Gonzalez and Narges Norouzi

**Reference Book Chapters:** Chapter 4.2 and 4.3